The Hawk in the Room:
Committee Composition and the Monetary Policy Shock

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This version: October 2026

Abstract

Monetary policy shocks are often measured as deviations from a policy rule that treats the central bank as a single decision-maker. Yet committees set policy, and their members often disagree. Using every individual vote cast on the Bank of England's Monetary Policy Committee since 1997, I construct a meeting-by-meeting measure of each member's hawkishness. Committee hawkishness, known before each meeting, predicts up to a fifth of the variation in the conventional UK monetary policy shock: part of that "exogenous" shock is the Committee's predictable response to its forecast. Removing this component eliminates the price puzzle in point estimates.

JEL Classification: E52, E58, D71, D72.
Keywords: Monetary policy, monetary policy committees, committee heterogeneity, constant-gain learning.

[1]Email: marina.koca.21@ucl.ac.uk. I thank my advisers, Morten O. Ravn and José-Víctor Ríos-Rull, for their invaluable guidance and support. For helpful comments I thank Mark Aguiar, Agustín Barboza Da Rosa, Michael Bordo, Piero De Dominicis, Yuriy Gorodnichenko, Daniel Lewis, Karthik Sastry, and Vincent Sterk as well as seminar and conference participants at the Bank of England brown-bag seminar, the Princeton University Macro Seminar, and the 18th South-Eastern European Economic Research Workshop. Platon Fakas provided excellent research assistance. I gratefully acknowledge financial support from the Onassis Foundation. All errors are my own.

1 Introduction

Members of the Bank of England's Monetary Policy Committee (MPC) deliberate over a common macroeconomic forecast, yet they regularly disagree in their policy votes. The disagreement is not random: some members vote persistently on the hawkish side of their colleagues and others on the dovish side, and those positions also evolve over a tenure. The equations economists estimate to describe how the Bank sets interest rates in response to the economy, known as policy rules, nonetheless typically treat the Committee as a single decision-maker. The monetary policy shocks identified from those rules inherit that assumption. This raises a simple question: does it matter where the Committee stands when it meets? If it does, a rule that omits the balance of views will record part of the Committee's systematic response as a policy shock. I find that it does. The balance of views changes the weight the rule places on the Bank's own forecast, and part of what the rule records as a shock is predictable from it.

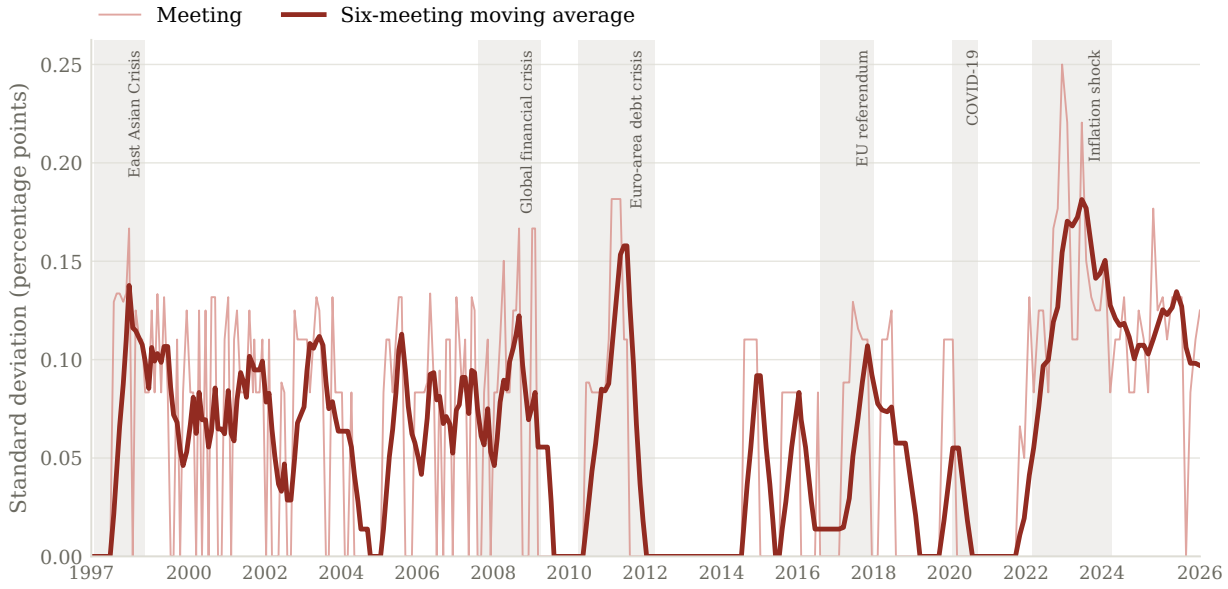


Figure 1: Dispersion of MPC Interest Rate Votes, 1997–2026. *Notes:* The figure plots the meeting cross-sectional standard deviation of members' preferred Bank Rate between June 1997 and July 2026 (light line) and its six-meeting moving average (dark line). Stretches of zero dispersion at the effective lower bound reflect unanimous rate votes when disagreement was expressed through asset-purchase votes. Shaded regions mark the labeled episodes of macroeconomic stress.

Figure 1 makes the scale of that disagreement concrete. The cross-sectional standard deviation of members' preferred rates reaches 0.25 percentage points at the December 2022

meeting, so dispersion is of the same order as a conventional 25-basis-point policy move. Dispersion of this order recurs throughout the sample rather than in a single episode, and its six-meeting moving average peaks at 0.18 percentage points in June 2023.

Whether such disagreement matters beyond the committee room depends on how it affects the policy rule. A policy rule, in the tradition of Taylor (1993), is an estimated equation that approximates how a central bank sets its interest rate in response to economic conditions; it summarizes the systematic pattern in past decisions rather than assuming that the central bank follows it mechanically. It splits each decision into a systematic part, the predictable response to the outlook, and a residual that the outlook cannot explain. An influential empirical literature treats that residual as an exogenous monetary policy shock (Christiano et al. 1999; Romer and Romer 2004; Cloyne and Hürtgen 2016), and such shocks are the basic input for measuring how monetary policy affects output and inflation. The interpretation requires the residual to be orthogonal to any information available when the decision is taken. Persistent hawkishness is exactly such information: if members carry leanings not reducible to the shared outlook, the balance of views on the Committee is known before the vote is cast, and what the rule records as a surprise is partly anticipated. This paper asks whether committee hawkishness affects how monetary policy is set and, as a result, the monetary policy shocks the literature uses to assess the effects of policy on the economy.

To guide the empirical analysis, I set out a stylized model of committee decision-making. Each member has a preferred interest rate that depends on the Bank's forecast, and a more hawkish member may want tighter policy across the board or may react more strongly to the forecast. The Committee's decision is a compromise between its members' preferred rates. In the model, that compromise depends on members' hawkishness only through its weighted average, so a single statistic of the Committee enters the policy rule. The model distinguishes two ways in which this statistic can enter. In the first, a level shift, a more hawkish Committee sets tighter policy whatever the outlook. A conventional rule then captures the response to the forecast correctly but misses this shift. In the second, a change in responsiveness, a more hawkish Committee reacts more strongly to the same forecast, raising rates by more when inflation is projected above target. A conventional rule, which imposes the same response at every meeting, then gets the response itself wrong. The Bank's explicit inflation target allows the data to distinguish the two. A level shift makes the Committee tighter even when inflation is forecast to be at target. A change in responsiveness affects policy only when the forecast departs from the target. The model

also defines the policy shock precisely: the part of the decision that cannot be predicted from the information the Committee holds on entering the meeting, including its balance of views.

Answering this question requires measuring the Committee's hawkishness, which is not ordinarily observable and is typically imputed from the assessments of professional observers (Istrefi 2019; Bordo and Istrefi 2023). The Bank of England removes that obstacle. Unlike the Federal Open Market Committee (FOMC), whose members vote for or against a proposed directive, each MPC member votes for a level of Bank Rate, and the Bank has published every vote by name at every meeting since operational independence in 1997, including votes on asset purchases while Bank Rate was at its effective lower bound. A vote reveals a member's hawkishness only when it is compared with the economic conditions at the time: a vote for a 25-basis-point increase is hawkish when inflation is projected below target, but is what a standard policy rule would prescribe when inflation is projected well above it. I therefore classify each vote against a common benchmark: the rate change implied by a forward-looking policy rule in the tradition of Romer and Romer (2004), estimated on the Bank's real-time forecasts as in Cloyne and Hürtgen (2016), whose UK approach I extend through 2026.[2] The residual of this rule, which I call the baseline shock, is the conventional UK monetary policy shock that the paper examines. Because the benchmark is defined at every meeting, the classified votes yield a time-varying measure of each member's hawkishness, which I call vote-implied hawkishness. I update each member's score after every vote with a constant-gain filter, as in the adaptive-learning literature (Evans and Honkapohja 2001): each new vote moves the score a fixed fraction of the way toward it, so older votes are gradually discounted and the score keeps moving throughout a member's tenure. The asset-purchase votes carry the measure across the lower-bound years, giving a continuous series from 1997 to 2026. The measure captures revealed hawkishness, which may reflect preferences, beliefs, or private information. The paper does not attempt to separate these sources: the question is whether the Committee's hawkishness belongs in the systematic policy rule.

Before turning to the policy rule, the member-level measure reveals two patterns. First, members differ persistently: a member whose votes are classified as more hawkish than their colleagues' early in a tenure tends to remain so later, with a correlation of 0.57

[2]The Committee's own decision cannot serve as the benchmark, because the members being measured set it. Against that benchmark, hawkishness would only be relative: a member would look more dovish simply because more hawkish colleagues joined, and a Committee that was hawkish as a whole would not show up as such.

between the two halves of a tenure across 49 members. Second, the Committee's average hawkishness changes even when its composition does not. A member joins or leaves the Committee at only 47 of 313 meetings. At the remaining meetings, the membership is the same as at the previous one, and average hawkishness moves because sitting members cast new votes that revise their scores. Turnover matters only episodically, although its effects accumulate over longer horizons.

The first result is that the Committee's balance of views belongs in the policy rule. Adding the average hawkishness of the Committee entering the meeting raises the share of policy decisions the rule explains by 20 percent and lowers its out-of-sample forecast error by 25 percent. Hawkishness matters mainly through the response to the forecast: the point estimates indicate that a more hawkish Committee raises rates by more when inflation is projected above target, as the model predicts. How divided the Committee is also matters. For a given average, a Committee whose members' views are more widely spread sets looser policy than the rule predicts, even after accounting for dissents at the previous meeting. Policy therefore depends on the whole distribution of members' views, not only on its center, which a single representative policymaker cannot reproduce.

This has a direct implication for the measurement of monetary policy shocks. The conventional shock, the residual from a rule that ignores the Committee's balance of views, is partly predictable from that balance before the meeting begins: committee hawkishness accounts for up to a fifth of its variation, mostly by shifting the response to the forecast. Because this information is available before the vote, the predictable part, which I call the committee component, is not a policy surprise but systematic policy that a rule with fixed coefficients cannot represent. The same pattern appears in surprises measured in financial markets. In the full sample, committee hawkishness accounts for 23 percent of the variation in the Target factor of the UK Monetary Policy Event-Study Database (Braun et al. 2025, 2026) that macroeconomic information leaves unexplained. That factor is built from intraday market prices and uses neither the votes nor the rule. Removing the committee component yields an adjusted shock that remains highly correlated with the conventional one, at 0.83, so the adjustment refines the conventional measure rather than replacing it.

The heterogeneity the votes reveal is not driven by measuring votes against a policy rule. The terms that let hawkishness shift the response to the forecast remain significant against a placebo that replaces the votes with the rule's own past residuals, and when each vote is compared with the Committee's decision instead of the rule. As a further check, I construct a second index of committee heterogeneity, which I call the formative-experience index,

that uses no votes at all, based on the macroeconomic conditions members experienced in their impressionable years. Beliefs about the economy formed early in life tend to persist (Krosnick and Alwin 1989; Malmendier and Nagel 2016), including among central bankers (Malmendier et al. 2021), and MPC members came of age in very different conditions: the stop-go cycles of the 1950s and 1960s, the inflation and stagflation of the 1970s, and the recessions that followed, with members raised abroad exposed to their own countries' conditions. Graduate training, which Bordo and Istrefi (2023) link to the types of FOMC members, varies little across MPC members, so the index uses the conditions members experienced between ages 10 and 17, decades before their appointment. Like vote-implied hawkishness, the index changes how strongly the rule responds to the Bank's inflation forecast, and it remains significant when vote-implied hawkishness is also included. This is further evidence that committee heterogeneity is a systematic feature of policy, one that the measurement of monetary policy shocks needs to account for.

Removing the committee component changes the estimated transmission of policy. Price and output puzzles are a recurrent feature of estimated responses to monetary policy shocks, and recent work traces them to information and predictability in the measured shock (Miranda-Agrippino and Ricco 2021; Bauer and Swanson 2023; Ricco and Savini 2025); the baseline UK shock displays both. With responses normalized so that Bank Rate rises by 100 basis points at its peak, the baseline price level rises between 17 and 23 months after a tightening before 2020, peaking at 0.11 percent, whereas under the adjusted shock it stays at or below zero at every horizon and is 0.13 percent below its starting level at 18 months. In the full sample, where the baseline puzzle reaches 0.21 percent at 18 months, the simplest adjustment, which uses only the level and change of average hawkishness, removes it. A decomposition of the baseline response shows why: most of that price increase comes from the committee component rather than from the adjusted shock. The conventional shock thus labels part of the Committee's systematic response to its inflation forecast as a surprise, the mechanism Sims (1992) proposed as the source of the price puzzle. Real GDP behaves differently. It still rises for about a year after a measured tightening under both shocks, and the decomposition shows that this expansion comes from the adjusted shock rather than from the committee component; the adjustment instead deepens the later contraction, to a trough of -0.95 percent at 31 months against -0.82 percent under the baseline. Omitted committee hawkishness therefore helps explain the price puzzle in the point estimates, but not the output puzzle.

The paper makes three contributions. First, it measures committee hawkishness directly

from members' votes, for every member and at every meeting. The measure shows that members hold persistent but evolving positions, so that the Committee's hawkishness moves even when its membership does not. Second, it shows that this committee state belongs in the systematic policy rule. Earlier work uses MPC votes to estimate member reaction functions (Besley et al. 2008), predict subsequent decisions (Gerlach-Kristen 2004), or identify the decision protocol (Riboni and Ruge-Murcia 2010); here the votes are aggregated into a predetermined state that representative-policymaker rules omit. Both the average and the dispersion of members' views matter, and the average matters mainly by changing the response to the Bank's forecast. Third, it shows that conventional monetary policy shocks absorb this systematic component. This complements Hack et al. (2025), who show for the FOMC that time variation in systematic policy contaminates conventional shocks, using perceived hawk–dove types and the rotation of voting rights, which shifts the committee's composition for reasons unrelated to the economy. Observed votes add variation within a fixed membership and allow the predictability to be traced to the votes themselves; the evidence also extends to market-based surprises and helps explain the UK price puzzle in the point estimates. Where policy is made by committee, the balance of views entering the meeting belongs to the systematic rule rather than to the shock.

Related literature. This paper joins three literatures. The first identifies monetary policy rules and shocks by specifying a reaction function and treating what it leaves unexplained as the innovation, whether as one equation of an identified system (Sims 1992; Bernanke and Blinder 1992; Christiano et al. 1999) or as a single equation conditioned on the central bank's own real-time forecasts (Romer and Romer 2004), which Cloyne and Hürtgen (2016) construct for the United Kingdom. It is that single-equation residual this paper takes as its object. A growing literature shows that measured policy surprises, including high-frequency ones, are predictable from information available before the decision (Miranda-Agrippino and Ricco 2021; Bauer and Swanson 2023), and that predictable components may reflect central-bank information or revisions to the perceived policy rule as well as policy shocks (Ricco and Savini 2025). This paper identifies one systematic, pre-meeting source of predictability in the forecast-residual shock and shows that it predicts part of a market-based surprise, without taking a stance on what that predictable component represents.

The second studies heterogeneity inside policy committees. That policymaker type is persistent and consequential is the premise of the delegation literature (Barro and Gordon 1983; Rogoff 1985); in a committee it becomes a question of aggregation. Riboni

and Ruge-Murcia (2008) show that the decision protocol shapes how individual preferences map into collective outcomes and generates endogenous inertia, and Riboni and Ruge-Murcia (2010) find that a consensus model fits the decisions of several committees, including the MPC, better than simple majority. For the FOMC the balance of views must be inferred: Istrefi (2019) classifies members from the assessments of professional Fed watchers, and Bordo and Istrefi (2023) trace those perceived types to graduate-school lineage and to conditions at birth. The MPC's attributed votes make the behavior itself observable and have been used to document systematic differences between internal and external members (Gerlach-Kristen 2003a; Besley et al. 2008; Berk et al. 2010; Bhattacharjee and Holly 2010), within-group heterogeneity (Harris and Spencer 2009), ideal-point dispersion (Hix et al. 2010; Eijffinger et al. 2018), and drift over tenures (Hansen and McMahon 2008). These studies pursue estimands related to but distinct from this paper's: member-specific reaction coefficients (Besley et al. 2008), the predictive content of dissent for subsequent decisions (Gerlach-Kristen 2004), and the protocol mapping preferences into decisions (Riboni and Ruge-Murcia 2010). This paper does not re-estimate preferences, member reaction functions, or the decision protocol. It aggregates members' revealed hawkishness into a predetermined, meeting-frequency committee state and asks whether representative-policymaker rules, and the shocks measured from them, omit that state. Hack et al. (2025) establish the general consequences of omitting time-varying systematic policy for the FOMC. This paper complements their evidence: it uses directly observed votes, distinguishes a slope from a level using the explicit inflation target, and shows that in the United Kingdom the committee state moves mainly through members' own updating, with changes in membership mattering over longer horizons.

The third asks where such differences originate. Hansen et al. (2014) structurally separate asymmetric policy preferences from the precision of members' private assessments and find that both contribute to MPC disagreement, with private assessments playing an important quantitative role. The distinction disciplines what vote-implied hawkishness can claim: preference heterogeneity generates persistent member rankings, whereas belief heterogeneity generates context-dependent disagreement. The evidence below fits neither pole cleanly, since member positions persist but also evolve. Vote-implied hawkishness combines preferences, beliefs, and private information and does not separate them; Appendix E.1 compares it with the member-level estimates of Hansen et al. (2014) and Besley et al. (2008). A related literature finds that policymakers' own macroeconomic experiences shape their beliefs persistently (Alesina and Fuchs-Schündeln 2007; Malmendier and

Nagel 2011, 2016); Malmendier et al. (2021) show that FOMC members' lifetime inflation experiences predict their votes, speech tone, and projections, and that meeting-average experience enters the federal funds rate rule additively. The formative-experience index of Section 7, which uses neither votes nor policy outcomes, instead enters through the forecast channel, the same channel as vote-implied hawkishness.

The remainder of the paper is organized as follows. Section 2 describes the institutional setting and the data. Section 3 sets out a model of committee aggregation and defines the policy innovation. Section 4 constructs the hawkishness measure, documents member positions, and decomposes movements in the Committee's hawkishness. Section 5 places vote-implied hawkishness in the policy rule, Section 6 examines what it implies for measured monetary policy shocks, and Section 7 reports evidence from members' formative experiences. Section 8 traces the consequences for estimated transmission, and Section 9 concludes. All appendices are in the Supplemental Appendix.

2 Institutional Setting and Data

The Committee. The MPC was established when the Bank of England was granted operational independence in May 1997, with a mandate to maintain price stability (defined by a government-set inflation target of 2 percent) and, subject to that, to support the government's objectives for growth and employment. Throughout the sample, the Committee has comprised nine members: the Governor, who chairs it; three Deputy Governors; the Bank's Chief Economist; and four external members appointed by the Chancellor of the Exchequer for fixed terms, usually three years and renewable once. The internal–external split is deliberate, with externals expected to bring independent assessments and to challenge the internal consensus. It also shapes the turnover of the Committee: externals rotate off on fixed terms while internals typically serve longer, so composition changes episodically rather than continuously, which is the variation the decomposition in Section 4 isolates.

Votes. Decisions are taken by simple majority, each member holding one vote.[3] Individual votes are published and attributed by name, and members vote on the level of the instrument rather than for or against a proposal, so both the direction and the intensity of dissent are observable. Attribution extends to votes on the size of the asset purchase program, which is what makes revealed hawkishness observable over the years at the effective

[3]The Governor has no formal veto, although in the rare event of a tie the Chair has a casting vote.

lower bound. Members are publicly accountable for these votes, through appearances before the Treasury Select Committee and through the open letter the Governor must write when inflation deviates from the target by more than one percentage point, which supports reading a vote as a considered statement of a member's view rather than as cheap talk. Appendix A sets out the Committee's meeting calendar, publication schedule, and accountability arrangements in more detail.

2.1 Data

All data sources are cited in the references: MPC votes (Bank of England 2026b), the Bank's report-vintage forecasts (Bank of England 2026a; Cloyne and Hürtgen 2016), the UKMPD factors (Braun et al. 2026), the Wu–Xia shadow rate (Wu and Xia 2026), ONS and FRED series (Office for National Statistics 2026d,b,a,c; International Monetary Fund 2026; Baker et al. 2026), and the author-compiled replication data (Kotsa 2026). The Supplemental Appendix opens with a data availability statement.

Meetings and votes. The voting panel records every MPC policy decision from operational independence through 30 July 2026: 314 meetings and 2,785 attributed votes cast by 49 members.[4] Numerical rate votes are retained at their reported level. For 23 votes at nine meetings in 1998 that the published record reports only as an increase or decrease, I reconstruct the preferred rate from the alternatives described in the published minutes, assuming a 25-basis-point move from the pre-meeting rate where the minutes give only the direction (Appendix B). After the forecast and unemployment lags are formed, the reaction-function sample contains 312 meetings.

Forecast vintages. The policy benchmark uses the Bank's own forecasts of inflation and real GDP growth from 117 published forecast vintages. From each vintage I take four quarters for each variable, giving 936 observations. The panel extends the forecast series of Cloyne and Hürtgen (2016), which ends in 2007, through July 2026 using the Bank's *Inflation Reports*, *Monetary Policy Reports*, and official numerical workbooks, and reconstructs the remaining figures from official vintages.

Each MPC decision is matched to the latest forecast available by the meeting, and the four quarters are dated relative to it: the quarter just ended and horizons zero through two.

[4] Howard Davies and Charlotte Hogg, who cast two votes and one vote respectively, are excluded because each has fewer than three recorded votes.

Forecast revisions compare two successive vintages' forecasts for the same calendar quarter, so they record how the Bank's view of that quarter changed rather than a shift in horizon. The full-history rule uses all eight forecast levels and six revisions, three per variable, through horizon one. A horizon-two revision would need the previous vintage's forecast three quarters ahead, which lies outside the four quarters kept, so these revisions enter only specifications in which that earlier forecast is independently observed. Appendix B documents the source and conditioning assumption of each observation, the vintage calendar, precision flags, and an independent source check for 2005–07.

Macroeconomic controls and outcomes. Unemployment is the ONS unemployment rate (series MGSX), entered with two lags over consecutive MPC decisions and dated by its release. The monthly transmission analysis uses the ONS monthly GDP and CPI indices, Bank Rate, oil-price inflation, and the IMF commodity-price index.

Formative experiences. For the formative-experience evidence of Section 7, I also code the macroeconomic experience of each of the 49 members included in the analysis between ages 10 and 17. Appendix J documents the coding and summarizes the members' education, generation, and pre-MPC employment.

3 A Model of Committee Hawkishness and the Policy Rule

This section sets out a stylized model of how a committee of heterogeneous members produces a single policy decision. It motivates which statistic of the committee an estimated reaction function should contain and separates two hypotheses about how committee heterogeneity enters such a rule—as an omitted level or as a misspecified slope—which the rest of the paper tests against each other.

3.1 Members

Let z_t collect the conditioning variables available to the Committee at meeting t: the Bank's forecasts of inflation and real GDP growth, $\pi^F_{t,j}$ and $y^F_{t,j}$, at horizons $j = -1, \ldots, 2$ quarters, where $j = -1$ is the quarter just ended; their revisions; and lagged unemployment. Inflation forecasts enter as gaps from the target in force, $G_{\pi,t,j} = \pi^F_{t,j} - \pi^*_t$, where π^*_t is 2.5 percent on RPIX through the August 2003 forecast and 2 percent on CPI from November 2003. Revisions already have a natural zero, no change in the forecast. Activity forecasts

enter uncentered, because the United Kingdom has no official output objective to measure them against. Section 3.3 explains why this matters.

Member i's preferred change in the policy rate at meeting t is

$$\Delta i^*_{i,t} \;=\; \alpha + \beta' z_t \;+\; \theta_{i,t-1}\big(\gamma + \Lambda' z_t\big) \;+\; u_{i,t}, \tag{1}$$

where $\theta_{i,t-1} \in [0,1]$ is the member's hawkishness entering the meeting, the probability of voting for a tighter setting than a common policy benchmark, as recovered in Section 4, and $u_{i,t}$ is idiosyncratic. The hawkishness is dated $t-1$ throughout: it is the position a member holds on arriving at the meeting, so it is predetermined with respect to the decision taken there, and it is the object the empirical specifications use. Writing the type as a scalar multiplying a vector of responses separates two distinct roles. The scalar γ is an *additive* hawkishness effect: how much tighter a hawkish member wants policy when every centered conditioning variable is at zero, that is, with inflation at target. The vector Λ governs how hawkishness changes the *weight* the member places on each element of z_t.

3.2 Aggregation

Suppose the Committee reconciles members' preferred settings by minimizing a weighted quadratic loss,[5] with nonnegative weights $w_{i,t}$ summing to one,

$$\Delta i^c_t \;=\; \arg\min_x \sum_i w_{i,t}\big(x - \Delta i^*_{i,t}\big)^2 \;=\; \sum_i w_{i,t}\,\Delta i^*_{i,t},$$

which gives

$$\Delta i^c_t \;=\; \alpha + \beta' z_t + C_{t-1}\big(\gamma + \Lambda' z_t\big) + u_t, \qquad C_{t-1} \equiv \sum_i w_{i,t}\,\theta_{i,t-1}. \tag{2}$$

where $u_t \equiv \sum_i w_{i,t} u_{i,t}$.

The reaction function therefore contains exactly one statistic of the committee, C_{t-1},[6] and the weights determine which: equal weights $w_{i,t} = 1/N_t$ give the committee mean

[5] The quadratic form is a convenient way to obtain a weighted average and is not essential for the empirical analysis. An absolute-value loss, for example, would select the weighted median of members' preferred settings. Because the empirical exercise compares the mean, the median member, and the Governor directly (Appendix D.4), the choice of summary does not rest on this assumption. The implication that dispersion drops out of the rule does rely on it, and Section 5 tests it.

[6] The empirical specifications also include the change in hawkishness, ΔH_{t-1}. It does not follow from this static aggregation and enters as a dynamic control.

$\bar{H}_{t-1}$; weight concentrated on the median member gives the median $\theta_{i,t-1}$; weight concentrated on the chair gives the Governor's $\theta_{i,t-1}$.

The model does not pin down the weights, and the empirical exercise does not attempt to identify them. Formal majority voting points to the median member, whereas consensus-seeking points toward the mean, which Riboni and Ruge-Murcia (2010) find fits the decisions of several committees, including the MPC, better than simple majority. Appendix D.4 estimates the rule with each candidate in turn; the mean fits best on adjusted R^2 and BIC, and I use it throughout. I treat this as selecting the best reduced-form summary of the Committee's center, not as identifying its decision protocol. Because Equation (2) is linear in θ, the dispersion of views drops out of the rule, so evidence that dispersion matters is evidence against linear aggregation. Section 5 reports that evidence, and Appendix H sets out two departures that give dispersion a role.

Policy innovations. Equation (2) also makes precise what counts as a monetary policy shock. Let $\mathcal{I}_t^{CB}$ denote the information the Committee holds when it meets at t: the outlook z_t and the balance of views C_{t-1} it brings to the meeting. I define the monetary policy innovation as the part of the decision that cannot be predicted from that information,

$$\varepsilon_t^{MP} \equiv \Delta i_t^c - E\left[\Delta i_t^c \,\middle|\, \mathcal{I}_t^{CB}\right], \tag{3}$$

that is, the realized change in the policy rate minus the change expected given the Committee's information. Under Equation (2) this innovation is u_t, and everything else, including the contribution of C_{t-1}, is systematic policy. The classification rests on information and timing alone: committee hawkishness is systematic because it is known before the decision, not because it is assumed to arise from non-monetary forces. Nor does the definition remove all hawkishness-related variation. A change in hawkishness that occurs at meeting t itself is not part of C_{t-1}, stays in the innovation, and may be a genuine policy shock. The empirical question is therefore whether a rule with fixed coefficients and a single representative policymaker omits this predetermined state, not where that state ultimately comes from.

3.3 Additive Omission versus Slope Misspecification

Equation (2) nests two views of how committee heterogeneity enters an estimated reaction function. Under *additive omission*, $\gamma \neq 0$ and $\Lambda = 0$: the conventional rule has the right slopes and omits a level. This is the form in which committee and experience-based

measures have entered estimated policy rules, as in Malmendier et al. (2021). Under *slope misspecification*, $\Lambda \neq 0$: the rule's response to the outlook varies with the Committee's composition, so a fixed-coefficient rule is misspecified rather than merely incomplete. This is the form in which Hack et al. (2025) model time-varying systematic policy, with the FOMC's hawk–dove balance shifting the rule's slopes.[7]

As with any interaction model, the level coefficient depends on where the interacted variables are centered. For an element x_t with mean m,

$$C_t\, x_t \;=\; C_t\,(x_t - m) \;+\; m\, C_t, \tag{4}$$

so a pure slope-misspecification process estimated on uncentered regressors also produces a level term, with coefficient $m\Lambda_x$. This is why inflation forecasts enter as gaps from the target (Section 3.1): γ is then the hawkishness effect with inflation at target, and Λ_π the change in the response to an inflation gap, which is positive if more hawkish members place more weight on inflation stabilization. An RPIX-regime indicator and its interaction with hawkishness absorb the 2003 change in the target.[8] The activity forecasts have no such zero, so their interactions are descriptive, and when they are included the level coefficient is not a clean estimate of γ. Appendix C derives what each hypothesis implies for the residual of the estimated rule.

4 Measuring Committee Hawkishness

Section 3 motivates the object the policy rule should contain: the committee-weighted average of members' latent hawkishness entering the meeting. This section recovers it. I first set out the latent-hawkishness framework and the prior, then describe how votes are translated into informative signals in normal times and at the effective lower bound, and what the construction identifies. I then present the estimated series and its sensitivity, and decompose its movement into within-member revision and turnover.

[7] They take the rule's inputs to be mean zero and note, but abstract from, a formulation in which inputs are measured relative to target. The Bank's inflation target is explicit and changed once, in 2003, so I adopt that formulation here, which fixes where the level term is evaluated.

[8] Writing $\pi_t^* = 2 + 0.5 D_t^{\text{RPIX}}$, $C_{t-1}\pi_t^F = C_{t-1}(\pi_t^F - \pi_t^*) + 2\,C_{t-1} + 0.5\,C_{t-1}D_t^{\text{RPIX}}$. The fitted values of the two parameterizations agree to numerical precision.

4.1 Hawkishness, Priors, and Voting Signals

I model each MPC member i as having a latent hawkish probability $\theta_{i,t} \in [0,1]$: the probability of taking a position tighter than the policy benchmark at meeting t. The probability captures a revealed tendency toward tighter versus looser policy after conditioning on the benchmark. It may reflect preferences, beliefs, or private information, and is treated as slowly varying rather than fixed, consistent with fixed-term appointments and a stable mandate: members may re-evaluate their position over a tenure, but do so gradually relative to the frequency of meetings. I refer to its estimate, $\hat{\theta}_{i,t}$, as the member's vote-implied hawkishness, or hawkishness score; a member is classified as hawkish at meeting t when the score exceeds one half.

It is useful to think of $\theta_{i,t}$ as having two parts,

$$\theta_{i,t} = \mu_i + \delta_{i,t}, \tag{5}$$

where μ_i is a persistent component specific to the member and $\delta_{i,t}$ captures state-dependent movement around it. The filter does not separately identify these components: the object recovered at each meeting is their sum. Tenure averages and the shift-share decomposition are therefore descriptive ways of distinguishing persistent cross-member differences from within-member updating, not estimates of structural types. Section 7 instead uses members' formative experiences, fixed before they joined the Committee, as a proxy for one possible source of the persistent component.

I observe a sequence of voting decisions over a member's tenure, $\{V_{i,1}, \ldots, V_{i,T_i}\}$. Votes are interpreted as noisy signals of $\theta_{i,t}$, drawn as

$$V_{i,t} \mid \theta_{i,t} \sim \text{Bernoulli}(\theta_{i,t}), \tag{6}$$

so that a member with a higher $\theta_{i,t}$ is more likely, but not certain, to vote on the tight side of the benchmark at any given meeting. The object of inference is $\theta_{i,t}$, tracked meeting by meeting as the voting record accumulates.

The initial prior follows a Beta distribution:

$$\theta_{i,0} \sim \text{Beta}(\alpha_0, \beta_0), \tag{7}$$

with prior mean $\pi_{\text{prior}} = \alpha_0/(\alpha_0 + \beta_0)$ and prior strength $s = \alpha_0 + \beta_0$, the latter interpreted as an effective sample size. In the baseline every member begins their tenure at the neutral value $\pi_{\text{prior}} = 0.5$, so that classification is driven by the voting record rather than by any presumption about the member, and $s = 15$.

Votes are not directly comparable across meetings because economic conditions vary over time. I therefore translate votes into binary signals relative to economic fundamentals using a Taylor-rule benchmark. Let $v_{i,t}$ denote member i's preferred policy rate at meeting t, and define the member's preferred change $\Delta v_{i,t} \equiv v_{i,t} - i_{t-1}$, where i_{t-1} is the policy rate inherited from the previous meeting. Let τ_t^* denote the change in the policy rate implied by the forward-looking Taylor rule of Equation (12), that is, its fitted value at meeting t. Both objects are therefore measured as changes. I define the deviation

$$d_{i,t} = \Delta v_{i,t} - \tau_t^*, \tag{8}$$

and classify votes as

$$V_{i,t} = \begin{cases} 1 & \text{if } d_{i,t} > 0 \quad \text{(hawkish)}, \\ 0 & \text{if } d_{i,t} \leq 0 \quad \text{(dovish)}. \end{cases} \tag{9}$$

4.2 Constant-Gain Updating from Votes

Beliefs about $\theta_{i,t}$ are revised meeting by meeting as votes are observed. Because $\theta_{i,t}$ is allowed to drift, inference takes the form of a filtering problem: each new vote is informative not only about where the member has stood historically but about where they stand now. Combining the Beta prior with the Bernoulli signal in Equation (6) and re-centering the prior on the previous meeting's estimate delivers the recursion

$$\hat{\theta}_{i,t} = \hat{\theta}_{i,t-1} + g\left(V_{i,t} - \hat{\theta}_{i,t-1}\right), \qquad g = \frac{1}{s+1}, \tag{10}$$

with $\hat{\theta}_{i,0} = \pi_{\text{prior}}$. Each vote moves the estimate a fixed fraction g of the way toward the signal, so beliefs respond to new information at a constant rate.[9]

Equation (10) is the constant-gain learning rule of the adaptive-learning literature (Evans and Honkapohja 2001; Malmendier and Nagel 2016), and it is built to a specific requirement. The object of interest is where a member stands at a given meeting, and that position must remain estimable at meeting frequency for a member, regardless of their tenure. A constant gain holds the responsiveness to each new vote fixed at g, so

[9]The Beta–Bernoulli formulation keeps the score on the unit interval and yields a transparent one-step update. Because the prior is re-centered on the previous estimate with fixed strength, the procedure is a constant-gain filter, equivalent to an exponentially weighted moving average of past voting signals, rather than full conjugate (accumulating) Bayesian updating.

the estimate remains capable of moving throughout a tenure.[10] What $\hat{\theta}_{i,t}$ measures is accordingly a member's recent voting position rather than their record as a whole. At every meeting, the score is constructed recursively from that member's voting history up to that date.

The gain itself is calibrated through the Kalman correspondence: Equation (10) is the steady state of a filter for a local-level model with innovation variance q in $\theta_{i,t}$ and noise variance r in the vote, for which $q/r = g^2/(1-g)$. The baseline $s = 15$, or $g = 1/16$, therefore corresponds to a signal-to-noise ratio of approximately $1/240$: an estimate that is permitted to move, but slowly relative to the meeting-by-meeting noise in individual votes. The weight on a vote cast k meetings earlier declines geometrically at rate $(1-g)^k$, a half-life of roughly eleven meetings, or about a year and a half at the MPC's meeting frequency, so $\hat{\theta}_{i,t}$ is a geometrically discounted average that weights more recent behavior. Appendix D sets out the assumptions and the prior parameterization, and Section 4.3 reports the sensitivity of the measure to the prior and the gain.

The committee-level measure averages the scores of the members present at each meeting,

$$\bar{H}_t = \frac{1}{N_t} \sum_{i \in \mathcal{M}_t} \hat{\theta}_{i,t}, \tag{11}$$

where $\mathcal{M}_t$ denotes the set of members present at meeting t and $N_t = |\mathcal{M}_t|$ is the number of attending members. Since each $\hat{\theta}_{i,t}$ is the probability that member i votes on the tight side of the benchmark, $\bar{H}_t$ is the expected share of the Committee doing so at meeting t. I refer to $\bar{H}_t$ as committee hawkishness.

Normal period. For the period when the interest rate was not constrained by the effective lower bound (ELB), I use a forward-looking Taylor rule, the baseline rule, with a specification similar to those of Romer and Romer (2004) and, for the United Kingdom, Cloyne and Hürtgen (2016), in the spirit of the reaction-function estimates of Gerlach-Kristen (2003b):[11]

[10] If votes instead accumulated, with the weight on each new vote falling as $1/n$, the score of a long-serving member would become progressively immovable, removing meeting-frequency variation precisely where the voting record is longest.

[11] The rule is used as a forecasting device for the predicted change in the interest rate; it approximates central-bank rate setting closely without assuming that the MPC mechanically follows it.

$$\Delta i_t = \alpha + \sum_{j=-1}^{2} \phi_j \pi^F_{t,j} + \sum_{j=-1}^{2} \gamma_j y^F_{t,j} + \sum_{j=-1}^{1} \delta_j \left(\pi^F_{t,j} - \pi^F_{t-1,j}\right) + \sum_{j=-1}^{1} \kappa_j \left(y^F_{t,j} - y^F_{t-1,j}\right) + \sum_{j=1}^{2} \rho_j u_{t-j} + \varepsilon_t \tag{12}$$

Equation (12) specifies the policy benchmark used to translate individual voting decisions into relative monetary policy signals. The dependent variable, Δi_t, denotes the change in the policy rate between two consecutive MPC meetings, measured at a meeting-by-meeting frequency.

The superscript F denotes the forecast vintage available at the meeting, and $j \in \{-1, 0, 1, 2\}$ indexes the target quarter relative to that information set. Thus $\pi^F_{t,j}$ and $y^F_{t,j}$ are inflation and annualized real-GDP growth forecasts, with $j = -1$ referring to the quarter just ended. The rule contains eight forecast levels, six same-target revisions over $j \in \{-1, 0, 1\}$, and two meeting-frequency unemployment lags: sixteen information controls in total. Revisions stop at $j = 1$ because revising the two-quarter-ahead forecast requires the previous report's three-quarter-ahead forecast, which the forecast panel does not record consistently.

To allow policy to respond not only to levels but also to changes in expectations, the specification includes forecast revisions relative to the previous information set, given by $(\pi^F_{t,j} - \pi^F_{t-1,j})$ and $(y^F_{t,j} - y^F_{t-1,j})$. These terms capture new information arriving between meetings and proxy for surprises in expected inflation and real activity. Finally, recent economic conditions are controlled for through lags of the unemployment rate, u_{t-j}, for $j = 1, 2$. The inclusion of these lags captures short-run labor market dynamics that may influence policy decisions beyond the forecast outlook, as in Cloyne and Hürtgen (2016).

The forecast regressors are report-vintage data aligned to publication dates, so they reproduce the Bank's information set rather than revised outturns (Orphanides 2001; Coibion and Gorodnichenko 2012). This timing ensures that each fitted value uses only forecasts published before the decision. Whether the resulting residual is also orthogonal to committee information is the question of Section 6.

QE period. From March 2009 until July 2016 Bank Rate was held at 0.5 percent, but members continued to cast attributed votes on the overall size of the asset purchase program, and those votes show material disagreement. I use them to keep the member histories continuous through the lower-bound window, from the first attributed asset-

purchase vote in April 2009 to July 2016. The benchmark re-estimates Equation (12) over this window with the Wu–Xia shadow rate (Wu and Xia 2016) in place of Bank Rate. Because expansions in asset purchases lower the shadow rate, I reverse the sign of the fitted shadow-rate change and compare policy directions rather than quantities; Appendix D.5 gives the full construction. Let $v_{i,t}^{QE} = \text{sign}(\Delta q_{i,t})$, where $\Delta q_{i,t}$ is the change in the stock of asset purchases member i votes for, denote member i's preferred direction—contraction, hold, or expansion—and let $b_t^{QE} = -\text{sign}(\widehat{\Delta s}_t)$ denote the corresponding benchmark direction. The dimensionless deviation is $d_{i,t}^{QE} = v_{i,t}^{QE} - b_t^{QE}$, and votes are classified as

$$V_{i,t}^{QE} = \begin{cases} 1 & \text{if } d_{i,t}^{QE} < 0 \quad \text{(hawkish)}, \\ 0 & \text{if } d_{i,t}^{QE} \geq 0 \quad \text{(dovish)}. \end{cases} \tag{13}$$

The inequality runs the opposite way to Equation (9) because a tighter-than-benchmark choice is a smaller quantity of purchases; in both regimes the signal equals one when the member is tighter than the benchmark.

What the construction identifies. Identification comes from repeated member-level deviations from a common benchmark. Benchmarking does not remove every common condition: an omitted meeting-level policy component affects every member's classification alike, so what the filter recovers cleanly is a member's position relative to the others voting with them.

Votes are treated as conditionally independent signals given the latent probabilities and the benchmark. This is a filtering assumption, not a claim that members do not interact: common deliberation, private information, and correlated beliefs remain in the meeting component rather than in individual scores.[12]

4.3 The Evolution of Average Committee Hawkishness

The procedure above delivers, for each member at each meeting, an estimate of $\theta_{i,t}$, the probability of voting for a tighter setting than the benchmark prescribes. Averaging over the members voting at a meeting gives the committee-level hawkishness measure. For the descriptive series in this section, I use the rate-vote measure outside the effective-lower-bound period and the QE-vote measure within it.[13] The reaction-function analysis uses

[12] A richer model could represent strategic interaction or reputation explicitly; see, for example, Hansen and McMahon (2016).

[13] Bank Rate was also near its floor from March 2020 to December 2021, but it moved twice in that window, so the conventional benchmark remains estimable and I retain the rate-vote measure.

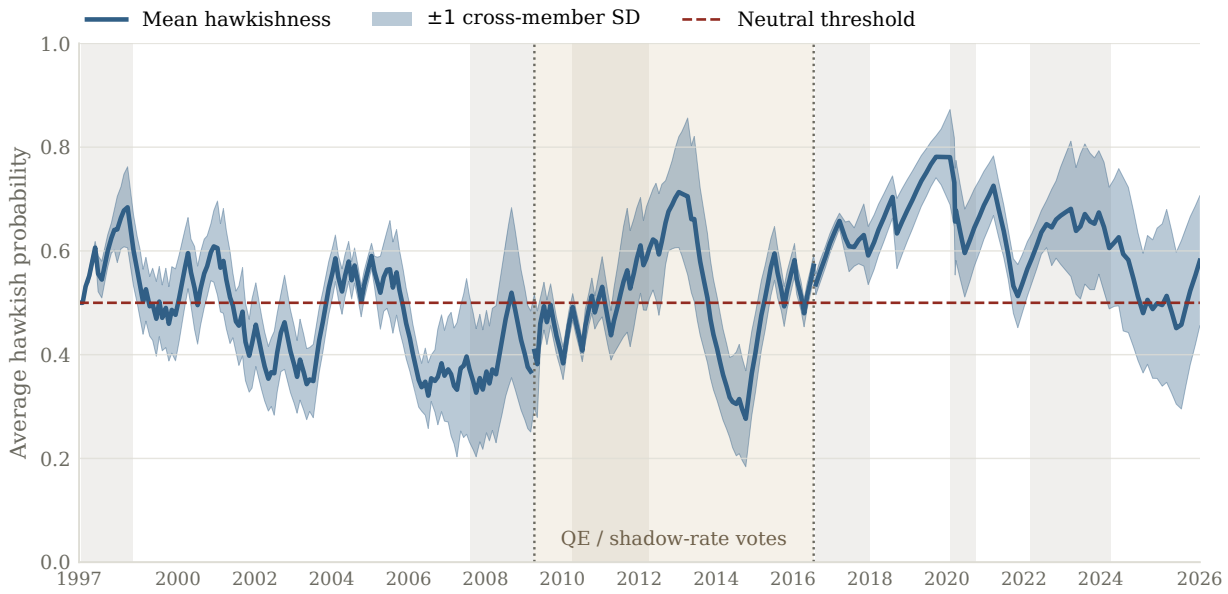


Figure 2: Average Vote-Implied Committee Hawkishness, 1997–2026. *Notes:* Hawkishness is measured from rate votes against the conventional benchmark, and between the dotted lines (April 2009–July 2016) from asset-purchase votes against the shadow-rate benchmark. Because the instrument and benchmark change at each boundary, levels are comparable only within segments. The blue band shows ±1 cross-sectional standard deviation; the red dashed line marks the neutral value of 0.5. Gray shading marks the stress episodes dated in Appendix B.

the measure appropriate to each policy instrument and reports the regimes separately. I begin with the two benchmark rules, then describe the resulting series and its sensitivity; Section 4.5 separates movements driven by sitting members from those driven by changes in committee composition.

The benchmark rules. A vote is informative only relative to the policy setting prescribed by the benchmark, so two rules are used. Panel A of Table 1 reports both coefficient vectors. When Bank Rate is the active instrument, the dependent variable is its meeting-to-meeting change and the information set is Equation (12). The inflation- and real-GDP-forecast level blocks are jointly significant under HAC(4) inference ($p < 0.001$ and $p = 0.048$, respectively). During the April 2009–July 2016 lower-bound window, the same rule, with the revision terms extended to $j = 2$, is estimated on the change in the Wu–Xia shadow rate (Wu and Xia 2016) over 88 meetings. As described above, only the direction of its fitted change is used to classify votes (Appendix D.5). Neither rule is interpreted structurally; each serves to supply a transparent conditional policy prescription whose adequacy can be challenged directly in the robustness exercises.

The series. Figure 2 plots the regime-combined committee average $\bar{H}_t$ over 1997–2026.[14] Because the instrument and classifier change over the lower-bound interval, levels are comparable only within each plotted segment. Within segments, the series moves for two distinct reasons. The first is individual dissent: $\bar{H}_t$ rises through 1998, when several members repeatedly voted for increases the majority rejected, and through the 2022–23 tightening, when members pressed for larger increases, reaching 0.68 in March 2023; it falls to 0.45 by December 2025 as dovish dissents accumulate during the easing cycle. The second is the Committee's collective position relative to the benchmark. The full-series maximum of 0.78 in September 2019 reflects unanimous holds while the benchmark prescribed easing; the March 2020 emergency cuts, which went beyond the benchmark, lowered $\bar{H}_t$ to 0.66; and the delay in tightening through 2021, when the Committee held while the benchmark called for increases, lowered it to 0.51. The first source distinguishes members from one another; the second is common to all members at a meeting, and Section 6 examines how it relates to the residual of the rule.

Across all 314 meetings, $\bar{H}_t$ averages 0.523 with a time-series standard deviation of 0.112, and cross-member dispersion, $\sigma_{H,t}$, averages 0.077 (Appendix Table S8). Appendix Figure S1 shows each member's history, and Appendix Table S9 ranks members on the first of these sources: each member's average score relative to the colleagues voting at the same meetings. For instance, Tim Besley, Andrew Sentance, and Catherine Mann are among the most hawkish relative to their colleagues, and Swati Dhingra, David Blanchflower, and Adam Posen among the most dovish.

Because each member's history runs continuously across both instruments, the measure also allows internal and external members to be compared at the same meetings over the full sample, a distinction emphasized by Besley et al. (2008). Internal members have a mean tenure-average score of 0.538, compared with 0.503 for external members. With meeting fixed effects, internal members are 3.3 percentage points more hawkish on the score ($p = 0.083$) and 5.0 percentage points on the hawkish vote signal ($p = 0.051$), in line with the insider–outsider pattern documented in the literature. The conventional-rate subsample gives estimates of the same sign (Appendix Table S10). External members also account for most of the Committee's dovish tail (Appendix E), and Appendix H examines

[14] The dispersion of estimates across members is summarized by the cross-sectional standard deviation,

$$\sigma_{H_t} \equiv \left(\frac{1}{|\mathcal{V}_t|} \sum_{j \in \mathcal{V}_t} \left(\hat{\theta}_{j,t} - \bar{H}_t \right)^2 \right)^{1/2}, \tag{14}$$

which is shown in Figure 2 as a shaded band around the mean.

how this dispersion of views enters the rule.

The lower-bound window. Between April 2009 and July 2016 the series is built from asset-purchase votes against the shadow-rate benchmark.[15] These votes are what make the measure continuous. Bank Rate did not move for seven years, so a rate-vote measure alone would leave a gap in the record of every member who served through those years, and no record at all for members, such as Adam Posen, whose entire tenure fell inside the window. With the asset-purchase votes, each member's history continues without interruption across both instrument boundaries, and every member's position on leaving the lower bound reflects the full voting record.

Descriptive comovement. Appendix Figures S2 and S3 place the regime-combined series beside activity, prices, policy, and uncertainty. The most direct meeting-frequency relationship is with policy changes: changes in hawkishness and Bank Rate have correlation 0.341. Correlations in levels are harder to interpret, since they span three measurement regimes and persistent series can appear strongly correlated even when they are unrelated. The figures are therefore descriptive context; the test against market surprises from the UK Monetary Policy Event-Study Database (UKMPD) of Braun et al. (2025) in Section 6 and the predetermined formative measure in Section 7 provide the independent validation exercises.

Sensitivity of the measure. The series does not depend on the filter's two free parameters or on its timing. I re-estimate every member history under prior means of 0.25, 0.50, and 0.75 and gains of $1/32$, $1/16$, $1/8$, and $1/4$; measure committee hawkishness both immediately before the meeting and one meeting earlier; rebuild the histories from a leave-one-meeting-out Taylor benchmark, so that no meeting determines its own classification; and, for the lower-bound segment, restart member histories at the instrument boundary. The qualitative conclusions hold throughout, and the leave-one-out histories are numerically indistinguishable from the baseline (Appendix L). The data also support the maintained gain. Estimated by Bernoulli maximum likelihood on the vote panel, $g = 0.088$ (95 percent interval $[0.063, 0.116]$), and the lower end of the interval is essentially the baseline $g = 1/16$ (Appendix D.3).

[15] Average hawkishness moves from 0.368 in March 2009 to 0.407 in April, and from 0.571 in July 2016 to 0.536 in August; these one-meeting changes partly reflect the change of instrument.

4.4 Member Positions: Persistent but Evolving

Behind the Committee average lie the individual histories (Appendix Figure S1). To test whether members hold persistent positions independently of the filter, I use the raw voting signals. For each member I average the signal relative to the contemporaneous committee mean, separately over the first and second halves of the tenure. Across the 49 members with at least twelve usable signals, the two halves have a correlation of 0.57 ($p < 0.001$; Spearman 0.37, $p = 0.010$), rising to 0.65 ($p < 0.001$; Spearman 0.41, $p = 0.006$) for the 44 members with at least 24 signals. Since the two halves use different votes, members occupy systematically different positions relative to their colleagues.

These positions also change. Nearly half, 47 percent, of the variance in members' meeting-centered scores is within members rather than between them, so a member's hawkishness is better described as evolving over a tenure than as a fixed type.

The measure complements existing member-level studies of the MPC, which pursue different and in some respects more ambitious objectives. Hansen et al. (2014) estimate a structural model of voting that separates members' policy preferences from the precision of their private information, a distinction a reduced-form measure cannot draw. They report their estimates for internal and external members as two groups; the vote-implied hawkishness agrees with them on the direction of the group difference, with internal members the more hawkish, and adds variation within each group and over time. Besley et al. (2008) estimate a reaction function for each member, allowing members to differ in their intercepts, smoothing, and responses to the Bank's forecasts. The vote-implied score is a single summary of a member's recent tendency to vote tighter than a common benchmark, so it need not rank members in the same way as any one of those coefficients, and in practice the rankings are not closely related. The approaches answer different questions: theirs characterize how individual members set policy, whereas this paper's measure is designed to be aggregated into a meeting-frequency measure of committee hawkishness. Appendix E.1 reports both comparisons.

4.5 Shift-Share Decomposition of Committee Hawkishness

Average hawkishness could change either because the filter revises incumbent members' positions or because the Committee's membership turns over. I separate these channels with a shift-share decomposition of movements in committee hawkishness across consecutive meetings.

Framework. Let $h_{i,t}$ denote the estimated probability that member i votes on the tight side of the benchmark at meeting t, and let $w_{i,t}$ be a weight equal to $1/N_t$ if member i votes at meeting t (and zero otherwise), where N_t is the number of voting members. Average committee hawkishness is defined as

$$\tilde{H}_t = \sum_i w_{i,t} h_{i,t}. \tag{15}$$

I consider changes in average hawkishness between consecutive meetings, $\Delta\tilde{H}_t = \tilde{H}_t - \tilde{H}_{t-1}$, and apply a standard shift-share decomposition:

$$\begin{aligned} \Delta\tilde{H}_t = & \underbrace{\sum_i (w_{i,t} - w_{i,t-1}) h_{i,t-1}}_{\text{Composition}} \\ & + \underbrace{\sum_i w_{i,t-1} (h_{i,t} - h_{i,t-1})}_{\text{Within}} \\ & + \underbrace{\sum_i (w_{i,t} - w_{i,t-1})(h_{i,t} - h_{i,t-1})}_{\text{Interaction}}. \end{aligned} \tag{16}$$

For an entering member, I set the unobserved lagged score equal to that member's first observed score, $h_{i,t-1} = h_{i,t}$; for an exiting member, I carry the final observed score forward, $h_{i,t} = h_{i,t-1}$. This convention assigns entry and exit to the composition component while setting the within change to zero for non-continuing members. The *composition* component therefore captures changes in aggregate MPC hawkishness arising from member turnover, entries and exits, holding individual estimates fixed at these reference values. The *within* component reflects changes driven by the evolution of the estimated probabilities of continuing members, holding committee composition constant. The *interaction* term measures the covariance between changes in weights and changes in estimated hawkishness and is quantitatively small throughout the sample.

Figure 3 presents the shift-share decomposition of changes in committee hawkishness into a within-member updating component and a composition effect. The within component is plotted in blue and the composition effect in red; an increase in the latter indicates that incoming members carry higher scores than those they replace, while a flat segment signals the absence of turnover.

The two channels operate at different time scales. At meeting frequency, movements in committee hawkishness come almost entirely from revisions to sitting members' positions: membership changes at only 47 of 313 transitions, and the within component varies about

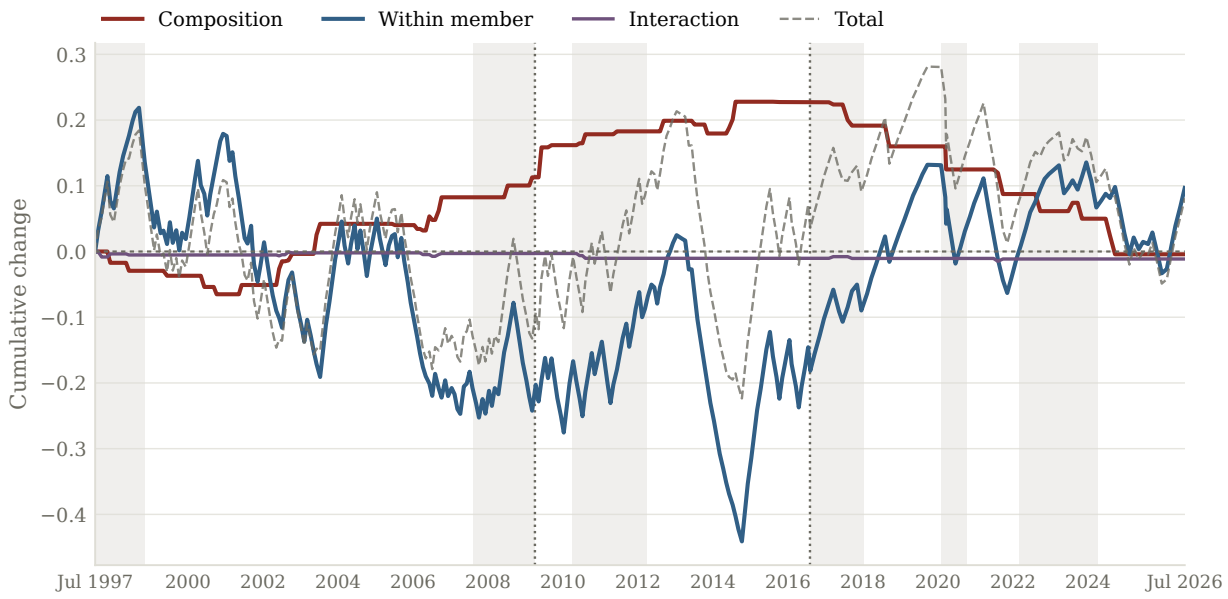


Figure 3: Shift-Share Decomposition of Cumulative Changes in Committee Hawkishness, 1997–2026. *Notes:* The red, blue, and purple lines report cumulative composition, within-member, and interaction effects; the gray dashed line is their sum. Gray shading marks the stress episodes dated in Appendix B. Vertical dotted lines bound the QE period (2009–2016), when hawkishness is inferred from asset-purchase votes relative to a shadow-rate benchmark.

four times as much as the composition component (standard deviations of 0.027 and 0.007).[16] Hawkishness moves substantially even when membership is fixed: no member joined or left between July 2003 and June 2005, or between August 2014 and August 2015, yet average hawkishness moved within a range of 0.20 and 0.32 over those stretches. Over longer horizons turnover matters more. It raised committee hawkishness by about 0.29 between 2000 and 2014, and later turnover largely reversed this between 2016 and 2024.

Persistent differences and within-member updating are therefore complementary features of MPC behavior. Members occupy persistently different positions relative to their colleagues (Section 4.4), and turnover changes which of those positions are represented on the Committee, which is why it matters over longer horizons. At meeting frequency, however, the Committee's hawkishness moves mainly because new votes by sitting members shift their estimated positions, whether because members change their views or because

[16]The constant gain of Equation (10) holds the responsiveness of the estimate to each new vote fixed at g, so the within component does not attenuate as a member's record lengthens, even though the weight carried by any individual past vote decays geometrically. The within component is a revision in the estimate, $h_{i,t} - h_{i,t-1} = g(V_{i,t} - h_{i,t-1})$, rather than an observed change in a member's view, so it is best read as movement in the Committee's position relative to the benchmark, holding composition fixed.

their votes reveal more about views they already held. These are statements about revealed voting positions; they do not separate changes in preferences from changes in beliefs or private information. Section 5 asks whether this information enters the policy rule.

5 Committee Hawkishness in the Policy Rule

If the balance of views inside the Committee carries information the forecast does not, it should show up in the rule the Committee's decisions trace out. Table 1 consolidates the main evidence on the policy rule. Panel A reports the complete coefficient vectors for the conventional Bank Rate and QE/lower-bound shadow-rate benchmarks, while Panel B presents the four-step committee-augmentation ladder in the full and pre-2020 samples. The two hawkishness terms are the standardized lagged committee level, H_{t-1}, and its lagged change, ΔH_{t-1}. Columns (2)–(4) of Panel B are three versions of the rule that differ in how these terms enter. The *level rule* adds them on their own, so that a more hawkish Committee can set tighter policy whatever the outlook. The *forecast-interaction rule* also lets them change the response to the inflation and GDP forecasts and their revisions. The *full-interaction rule* lets them change the response to every variable in the rule, and so measures the largest share of the residual that hawkishness can account for in sample. The shock adjusted using each rule is named after it. One common specification is used throughout, so the subsample conclusions are not selected separately. Appendix Table S13 reports every coefficient for the four specifications.

In the full sample the two hawkishness terms are jointly significant (HAC(4) $p = 0.002$) and raise adjusted R^2 by 20 percent.[17] A one-standard-deviation higher lagged level predicts a 1.86-basis-point larger subsequent rate change, and a one-standard-deviation larger lagged change predicts a 4.07-basis-point larger one. Before QE the two terms are not jointly significant ($p = 0.313$); in that period, as shown below, it is the dispersion of views that carries information. After the lower-bound period the level is the relevant margin: a one-standard-deviation higher level predicts a 4.4-basis-point larger rate change ($p = 0.011$), the change is insignificant, and the two terms are jointly significant ($p = 0.038$). In the lower-bound regime, the hawkishness terms are jointly significant in the shadow-rate equation ($p = 0.003$), with the level entering negatively (Appendix Table S13).

[17] Throughout, HAC(4) denotes Newey–West standard errors with a Bartlett kernel and four meeting lags. Four is the common rule-of-thumb lag length $T^{1/4}$ for the 312-meeting sample, and it comfortably covers the residual's persistence: with first-order autocorrelation of 0.40, the implied autocorrelation at the fourth lag is below 0.03.

Table 1: Policy-Rule Estimates and Committee-Hawkishness Augmentations

A. Benchmark rules

Regressor	Bank Rate	Shadow rate	Regressor	Bank Rate	Shadow rate
α	-0.154 (0.070)	0.416 (0.472)			
$\pi^F_{t,-1}$	-0.015 (0.021)	-0.164 (0.088)	$y^F_{t,-1}$	0.001 (0.002)	-0.008 (0.045)
$\pi^F_{t,0}$	0.118 (0.066)	0.452 (0.149)	$y^F_{t,0}$	0.019 (0.012)	0.084 (0.045)
$\pi^F_{t,1}$	-0.138 (0.094)	-0.316 (0.154)	$y^F_{t,1}$	-0.006 (0.010)	0.041 (0.072)
$\pi^F_{t,2}$	0.083 (0.055)	-0.014 (0.159)	$y^F_{t,2}$	0.032 (0.020)	-0.219 (0.118)
$\Delta\pi^F_{t,-1}$	0.061 (0.065)	0.920 (0.554)	$\Delta y^F_{t,-1}$	0.001 (0.005)	0.013 (0.062)
$\Delta\pi^F_{t,0}$	-0.154 (0.080)	-0.194 (0.371)	$\Delta y^F_{t,0}$	0.017 (0.012)	-0.033 (0.049)
$\Delta\pi^F_{t,1}$	0.101 (0.074)	0.290 (0.400)	$\Delta y^F_{t,1}$	0.053 (0.023)	0.089 (0.108)
$\Delta\pi^F_{t,2}$	—	-0.095 (0.252)	$\Delta y^F_{t,2}$	—	0.380 (0.163)
u_{t-1}	-0.055 (0.074)	-0.572 (0.382)	u_{t-2}	0.044 (0.073)	0.572 (0.353)
π^F levels: p-value	< 0.001	0.022	y^F levels: p-value	0.048	0.238
$\Delta\pi^F$ revisions: p-value	0.254	0.226	Δy^F revisions: p-value	0.056	0.014
u lags: p-value	0.332	0.202			
Adjusted R^2	0.294	0.220	Observations	312	88

B. Committee-hawkishness augmentations

	(1) Baseline	(2) Level	(3) Forecast interaction	(4) Full interaction
Full sample ($N = 312$)				
H_{t-1}	—	0.019 (0.009)	0.032 (0.011)	0.005 (0.012)
ΔH_{t-1}	—	0.041 (0.013)	0.037 (0.015)	0.034 (0.012)
Additive terms: p-value	—	0.002	< 0.001	0.003
Interactions: p-value	—	—	0.146	< 0.001
Adjusted R^2 (gain over baseline)	0.294	0.354 (+20%)	0.367 (+25%)	0.452 (+54%)
Pre-2020 ($N = 257$)				
H_{t-1}	—	0.012 (0.011)	0.011 (0.011)	-0.004 (0.012)
ΔH_{t-1}	—	0.027 (0.012)	0.027 (0.011)	0.025 (0.010)
Additive terms: p-value	—	0.028	0.019	0.050
Interactions: p-value	—	—	0.010	< 0.001
Adjusted R^2 (gain over baseline)	0.346	0.368 (+6%)	0.443 (+28%)	0.464 (+34%)

Notes: Parentheses contain Newey–West HAC(4) standard errors; all reported p-values use HAC(4) inference and jointly test the terms named in the row. Panel A uses the notation of Equation (12). The Bank Rate rule covers Aug. 1997–Jul. 2026 ($N = 312$); the shadow-rate rule covers Apr. 2009–Jul. 2016 ($N = 88$). Horizon-two revisions enter only the latter because independently observed numerical predecessors are available there.

In Panel B, column (2) adds standardized H_{t-1} and ΔH_{t-1}; column (3) interacts them with current inflation and GDP-growth forecasts and revisions; column (4) interacts them with all 16 information controls. Columns (2)–(4) are the level, forecast-interaction, and full-interaction rules and contain 2, 10, and 34 committee terms, respectively. A dash denotes a term or test that is not included.

Beyond the center. Among the candidate summaries of the Committee's center, the equal-weight mean fits best on adjusted R^2 and BIC, ahead of the median member and the Governor (Appendix Table S7). The dispersion of views around it, however, carries additional information. Adding the dispersion of member scores, measured relative to its largest possible value given the average, $D^{\text{rel}}_{t-1} = \sigma_{H,t-1} / \sqrt{\bar{H}_{t-1}(1-\bar{H}_{t-1})}$ (Equation 14; Appendix H), to the level rule raises adjusted R^2 from 0.354 to 0.384. The coefficient is negative: for a given average hawkishness, a more divided Committee chooses a smaller increase or a larger cut (HAC(4) $p < 0.001$). Dissent at the previous meeting is itself informative, as Gerlach-Kristen (2004) show, and the balance of hawkish over dovish dissents strongly predicts the next decision. Dispersion is not a restatement of that evidence: conditional on the number and direction of the previous meeting's dissents, it remains negative and significant (HAC(4) $p = 0.012$). The same sign holds before QE, when the level and change of average hawkishness are not jointly significant ($p = 0.015$). Dispersion is not significant within the lower-bound period, when only 20 of 88 meetings show any disagreement in asset-purchase votes, so that dispersion reflects accumulated histories rather than contemporaneous disagreement. Because linear aggregation implies that dispersion drops out, this is evidence that a representative-policymaker rule misses more than the Committee's center; Appendix H sets out two departures from linear aggregation that give dispersion a role. The shock adjustments below retain the mean, so that they correspond to the aggregation model of Section 3.

The two interaction rules let committee hawkishness change the response to the forecast itself. In the full sample they raise adjusted R^2 over the baseline by 25 and 54 percent, to 0.367 and 0.452, and in the pre-2020 sample (August 1997 to December 2019), which ends before the pandemic, by 28 and 34 percent. The full-interaction rule underlies the main pre-2020 impulse responses of Section 8.

I also ask whether committee hawkishness improves the rule's out-of-sample forecasts of policy decisions. I re-estimate the rule on an expanding window beginning in 2008 and forecast each subsequent decision.[18] The two hawkishness terms reduce the rule's mean squared forecast error by 25 percent in the full sample (Clark and West 2007 test, $p = 0.047$), and the forecast-interaction rule reduces it by 14 percent in the pre-2020 sample

[18]The rule's coefficients and standardization moments are re-estimated at every forecast origin; the historical hawkishness measure, and the benchmark used to classify the votes, are held fixed. The exercise therefore asks whether the measure adds predictive information when the rule is estimated recursively, not whether the measure itself could have been built in real time, which is a separate question. Appendix Table S14 reports the full grid and the forecasting design.

($p = 0.018$).[19] Committee hawkishness therefore improves the rule's predictions of policy decisions, not only its fit to past ones.

Because the full-interaction rule forecasts less well out of sample, I use it to bound the committee component and to decompose its effects rather than as the preferred forecasting model.

The model predicts that a more hawkish Committee responds more strongly to the inflation gap, the Bank's inflation forecast minus the target ($\Lambda_\pi > 0$). The full-sample estimates agree: the interaction of lagged committee hawkishness with the inflation gap is positive, both for the current-quarter forecast and for the average gap across the four forecast horizons (HAC(4) $p = 0.008$ and 0.005). When projected inflation is above target, a more hawkish Committee therefore raises rates by more, or cuts them by less. I treat this as suggestive: the four forecast horizons move closely together, which makes the interaction harder to pin down when they enter separately or when the sample is split (Appendix Table S3).

Observable confounds. A natural objection is that committee hawkishness proxies for a systematic variable that the benchmark omits. I add seven predetermined financial and uncertainty controls to the rule, first separately and then jointly: prior-month sterling depreciation, oil-price inflation, commodity-price growth, a corporate-spread proxy, broad UK equity-price growth, a financial-conditions proxy, and log economic policy uncertainty. Each expanded rule is used to reclassify every vote and rebuild every member history before the reconstructed measure is returned to that same rule, so the test is not conducted with the original index held fixed. On the balanced 285-meeting sample the reconstructed level is positive and significant in six of the seven single-control specifications; with all seven included jointly it remains 1.96 basis points per standard deviation ($p = 0.010$), close to the 1.86 basis points in the baseline, and the reconstructed level and change are jointly significant ($p = 0.001$). Committee hawkishness therefore carries information beyond the exchange-rate, commodity-price, financial, and uncertainty conditions that the benchmark omits. Appendix Table S24 reports the grid.

[19] The full-interaction rule has around fifty coefficients, too many to estimate precisely on the short early windows, and the post-2020 inflation episode lies outside the range those windows cover; I therefore use it in sample only, to measure how much of the residual the interaction terms can absorb.

6 Committee Information and the Measured Policy Shock

The residual of the baseline rule is predictable from committee hawkishness. Under the definition of Equation (3), the residual is therefore not the policy innovation, because the balance of views entering the meeting is part of the Committee's pre-decision information. This is what Section 3 implies if committee hawkishness belongs in the rule: under either hypothesis, the residual could inherit the part of the omitted committee terms that the rule's inputs do not capture (Equation S3). Entered on their own, the two hawkishness terms (the level block) have an adjusted R^2 of 0.081 in the baseline shock (raw R^2 0.087), and the full-interaction block, which contains all the committee terms of the full-interaction rule, has an adjusted R^2 of 0.208 (raw R^2 0.295). Both blocks are jointly significant (HAC(4) $p < 0.001$).

I next check whether this predictability reflects the voting record or some persistence of the residual itself. In particular, since each vote is classified against the estimated rule, each member's classified deviation decomposes into their dissent from the decision and the meeting-common residual:

$$d_{i,t} = \underbrace{(v_{i,t} - i_t)}_{\text{dissent from the decision}} + \underbrace{(\Delta i_t - \tau_t^*)}_{\varepsilon_t}. \tag{17}$$

Committee hawkishness could therefore carry part of that residual. I address this in four ways, which also bear on the two hypotheses of Section 3: under additive omission the level of committee hawkishness should carry the information, while under slope misspecification the interaction terms should.

The residual's own persistence. The most direct check conditions on the residual's own dynamics: three lags of the baseline shock and the second lag of the Bank Rate change (Table 2, Panel A).[20] Conditional on these dynamics, the level and change of hawkishness alone add nothing to the fit (adjusted R^2 0.177 against 0.178, HAC(4) $p = 0.477$). Adding the interaction terms, by contrast, raises the adjusted R^2 from 0.178 to 0.226, and the full-interaction block is jointly significant (HAC(4) $p < 0.001$).

The result does not depend on how many lags of the residual are included. With a single lag, the number BIC selects, the full-interaction block explains a further 16.8 percent of the variation the lags leave unexplained; with twelve lags it explains 18.4 percent. A more demanding check removes the residual's history from the measure itself by projecting

[20] The baseline shock is positively autocorrelated through three meeting lags (Appendix I.2).

the hawkishness level and change on up to forty lags of the residual. These projections absorb roughly half of their variation, yet the full-interaction block still explains a further 13.0 to 18.6 percent across projections on twelve, twenty, and forty lags. Because inference on the projected blocks depends on the covariance estimator, I base the formal test on the placebo comparison that follows.[21]

A residual-only placebo. The second check builds a placebo that shares the construction of the measure but contains no information from the votes. Every attending member receives the rule's continuous residual as their signal in place of a vote and updates at the same gain, $g = 1/16$, only at meetings they attend; entrants begin from the common prior; and the placebo is the mean across attending members. Its level, change, and interactions with all sixteen conditioning variables are formed exactly as for the actual measure, so the two blocks each contain 34 terms on identical support.

Conditional on the placebo and twelve lags of residual dynamics, the full-interaction block explains a further 21.7 percent of the variation left unexplained and is jointly significant under both HAC ($p < 0.001$) and conventional ($p = 0.008$) inference.[22] The level and change of hawkishness, by contrast, add only marginally, so the content that remains lies in the interaction terms. Because the placebo carries the residual's history, smoothed and interacted exactly as in the actual measure, this content can be attributed to the votes rather than to the residual's past.

A benchmark-free measure. The third check removes the common residual from the measure altogether. It enters the main measure because votes are classified against the rule; classifying each vote instead by its direction relative to the Committee's own decision, $+1$ above it, 0, or -1 below it, removes it by construction. I do not use this signed dissent measure as the main one because it captures hawkishness only relative to colleagues, but that is what makes it suited to this check. The signals are filtered with the same constant-gain rule and averaged across the serving Committee. I enter a six-term block: the level and change of the measure, their interactions with the current inflation and output-growth forecasts,

[21] The forty-lag projection absorbs 58.6 percent of the variation in the hawkishness level and 45.5 percent in its change. The HAC Wald tests on the full-interaction block reject at all three projection lengths ($p < 0.001$); the nested F tests give $p = 0.273$, 0.153, and 0.031 at twelve, twenty, and forty lags.

[22] Twelve lags cover more than one half-life of the filter at the maintained gain, and roughly one and a half at the gain estimated from the voting record (Appendix D.3). With three lags the HAC test still rejects ($p = 0.015$), but the conventional test does not ($p = 0.201$).

and the corresponding change-by-revision interactions. The block is jointly significant in the policy rule (HAC(4) $p < 0.001$ in the full and pre-2020 samples) and in the residual regression with dynamic controls ($p = 0.006$ and $p < 0.001$; Appendix Table S4). Although this check concerns predictability rather than transmission, it shows that the predictability result does not depend on classifying votes against the estimated rule.

Market-based policy surprises. The UKMPD Target factor (Braun et al. 2025, 2026) measures the policy-rate surprise in market prices around each meeting and is constructed from neither this paper's votes nor the baseline shock. I use it to ask whether committee hawkishness also predicts a surprise measured outside this paper's framework, not to validate the adjusted shock structurally. The predetermined level and change of hawkishness do not predict the factor ($p = 0.750$), whereas the full-interaction block explains an additional 23.0 percent of the variation left by the macro-information baseline and is jointly significant under HC3 inference ($p = 0.009$).[23] In the pre-2020 sample the block explains a similar 22.6 percent, although it is not statistically significant in the shorter sample ($p = 0.552$; Appendix I.3, which also reports the Path and QE factors).

This result establishes that the Target factor is not orthogonal to the predetermined full-interaction block. It does not identify the structural source of the predicted variation. The predicted component could represent systematic monetary policy, information about macroeconomic fundamentals, incomplete market conditioning on a committee-dependent reaction function, risk-premium variation, or measurement error (Ricco and Savini 2025). I therefore do not interpret the exercise as showing that every part predictable from committee hawkishness should be removed from the released factor.

Interpretation. Taken together, the tests locate the incremental predictive content primarily in the interaction terms. The additive level contributes little once the residual's own history is included, while the interaction terms survive the matched placebo and the benchmark-free voting measure and predict part of an independently constructed market surprise. For the baseline shock, this is the pattern Section 3 associates with slope misspecification: committee hawkishness changes how the rule responds to the forecasts rather than shifting its level. For the high-frequency factor, it is evidence of predictability rather than a

[23] I use HC3 rather than HAC(4) here because the surprise is measured in a narrow window around each announcement rather than accumulated from a persistent series such as the rule's residual, and because HC3 is the more conservative heteroskedasticity correction for a specification with 34 committee terms on about 300 meetings.

Table 2: Validation of Committee-Hawkishness Predictability

		Level block		Full-interaction block	
Exercise	N	Adj. R^2 [partial R^2]	p	Adj. R^2 [partial R^2]	p
A. Baseline-residual predictability					
No residual dynamics	309	0.081 [0.087]	< 0.001	0.208 [0.295]	< 0.001
3 residual lags + Δi_{t-2}	309	0.177 [0.004]	0.477	0.226 [0.164]	< 0.001
12 residual lags + Δi_{t-2}	300	0.211 [0.008]	0.205	0.269 [0.184]	< 0.001
B. Continuous-placebo tests					
Reaction function	312	0.412 [0.002]	0.815	—	—
Residual with 12 lags	300	—	—	0.393 [0.217]	< 0.001
C. UKMPD Target predictability					
Full released sample	302	-0.008 [0.003]	0.750	0.122 [0.230]	0.009
Pre-2020	253	-0.014 [0.005]	0.346	0.086 [0.226]	0.552

Notes: Fit cells report adjusted R^2, with partial R^2 relative to the row-specific parent in brackets. The level block contains lagged committee hawkishness and its change; the full-interaction block additionally contains their 32 interactions with the sixteen information controls. Panels A and B use HAC(4) inference. Panel B compares each actual block with its attendance-matched continuous-residual placebo. Panel C uses predictors lagged one meeting and HC3 inference.

structural decomposition: without separately observing market beliefs about the Bank's outlook and its reaction function, the exercise cannot distinguish central-bank information from imperfect market conditioning on committee hawkishness. Because the interactions involve persistent regressors, they may also capture some nonlinearity or gradual change in the policy rule. I construct adjusted shocks from both the level and the full-interaction rules; Section 8 reports the full-interaction shock in the main text and the appendix reports the level and forecast-interaction shocks.

Table 2 consolidates the residual-dynamics, matched-placebo, and UKMPD Target exercises. It reports fit and effect sizes as well as inference and keeps the full-sample external result beside its pre-2020 limitation.

7 Evidence from Members' Formative Experiences

Vote-implied hawkishness is recovered from the policy record itself. The placebo and benchmark-free checks of Section 6 show that its predictive content does not come from the rule's residual; a complementary question is whether committee heterogeneity is visible in data that contain no votes at all. If it is systematic, a source of it fixed long before any

member joined the Committee should enter the rule through the same forecast channel. This section asks whether it does.

The natural candidate for the Federal Reserve is graduate training: Bordo and Istrefi (2023) trace perceived hawk–dove types to graduate-school lineage. That variation is limited here. Of the 49 analysis members, 42 studied economics as undergraduates and 37 attended a "saltwater" institution, associated with the coastal Keynesian tradition rather than the "freshwater" tradition of Chicago and Minnesota. Moreover, Bordo et al. (2025) document that monetary economics was largely absent from British academic and policy training through much of the period in which MPC members were trained. What remains predetermined, observable, and plausibly formative is the macroeconomic environment members lived through before entering professional life. That is the variation Malmendier et al. (2021) exploit for the FOMC. I use the adolescent window of ages 10 to 17, chosen before examining policy outcomes and motivated by evidence on impressionable years (Krosnick and Alwin 1989; Blakemore and Mills 2014; Fuhrmann et al. 2015; Malave et al. 2022): for each member I summarize inflation and real-activity conditions over that window and aggregate to committee level by attendance. The index uses neither votes nor policy outcomes.

Conditional on vote-implied hawkishness, the formative-experience terms remain significant in a compact comparison with six interaction terms ($p = 0.022$). The signal is concentrated in changes in committee composition ($p = 0.012$) and in the response to the inflation forecast ($p = 0.023$). The index therefore captures a systematic part of committee heterogeneity from a source fixed decades before members' appointments.[24] Appendix J reports the member-level experiences, the formative-experience index and its evolution, and the full set of conditional tests.[25]

[24]The comparison is not symmetric: in the same specification, vote-implied hawkishness is not significant conditional on the index ($p = 0.326$), so the two measures overlap and neither dominates. The index is also correlated with birth cohort and policy regime, and its terms weaken to $p = 0.065$ when birth-cohort, policy-regime, governor, appointment-cohort, tenure, and internal-share interactions enter jointly. I therefore treat it as independent evidence that committee heterogeneity is systematic, not as a causal effect of formative years.

[25]A speech-sentiment index built from members' public speeches, following Firrell and Reinold (2020) and entered through the same interaction terms, is significant under HC1 ($p = 0.023$) and HAC ($p = 0.039$) inference but not under a block bootstrap, either alone ($p = 0.670$) or conditional on vote-implied hawkishness ($p = 0.821$). I do not use it further (Appendix Table S23). At the member level, the tone of a member's speeches does not predict their next vote once member and meeting effects are removed (Appendix L).

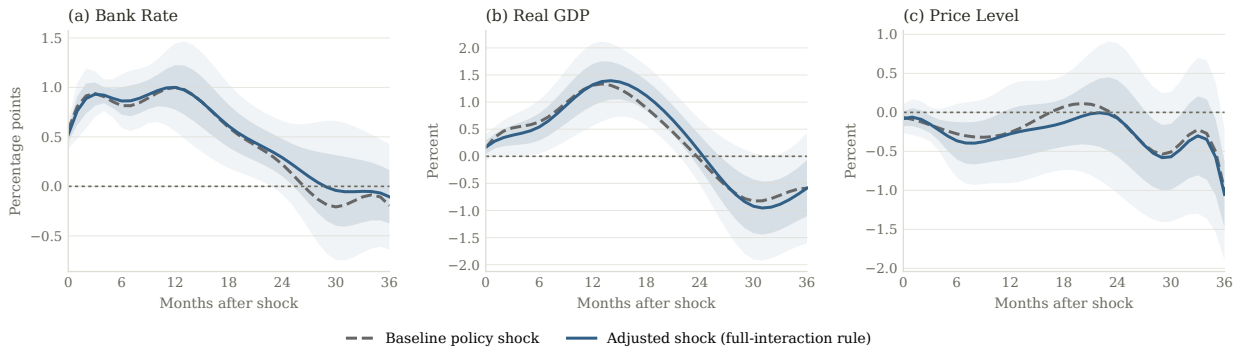


Figure 4: Impulse Responses to the Baseline and Full-Interaction Adjusted Shocks. *Notes:* Sample August 1997–December 2019. The adjusted rule adds lagged hawkishness, its change, and their interactions with the sixteen forecast, revision, and unemployment controls. Smooth local projections (nine-basis cubic B-spline, horizons 0–36), each scaled to a 100-basis-point peak Bank Rate response. All projections control for 12 lags of the Bank Rate, inflation, and log real GDP and a linear trend; the price projection also controls for 12 lags of oil-price inflation and the commodity-price index. Shaded regions are 68 and 95 percent pointwise bands for the adjusted response (Newey–West, 36 lags); baseline bands are omitted for clarity.

8 Implications for Estimated Transmission

Predictability matters for identification; the question here is whether it also moves estimated transmission. I construct the adjusted shock by re-estimating the baseline rule with the committee terms, each measured at one meeting's lag, so that every committee term is predetermined with respect to the residual being adjusted. The baseline and adjusted shocks are estimated separately in the full and pre-2020 windows on their common meeting support, aggregated to monthly frequency, and used in smooth local projections (Jordà 2005; Barnichon and Brownlees 2019) over 36 months; for the pre-2020 exercise, both shocks and outcomes end in December 2019. Responses are scaled to a 100-basis-point peak Bank Rate response. The main text reports the pre-2020 full-interaction adjustment, which implements the full-interaction rule of Section 6 on the pre-2020 sample; the paths for the level adjustment and for the full sample are reported in Appendix K.2. I report the full-interaction adjustment in the main text because it removes the largest committee component and so shows the largest change the adjustment can produce; the level and forecast-interaction adjustments are less demanding versions of the same exercise. Re-estimating the rule with these terms removes 29.7 percent of the variance of the pre-2020 baseline shock, an in-sample share rather than the adjusted R^2 of Section 6, and the adjusted shock retains a correlation of 0.84 with the baseline (Appendix I reports all specifications).

The price level. The clearest change is in the shape of the price response. Under the baseline shock the price level rises between 17 and 23 months after a tightening, peaking at 0.11 percent at 20 months. Under the adjusted shock the price level stays at or below zero at every horizon and is 0.13 percent below its starting level at 18 months, so, in the point estimates, the price puzzle disappears. The same pattern holds under the level adjustment. In the full sample, where the baseline puzzle is larger (0.21 percent at 18 months), the level adjustment removes it; in the pre-2020 sample it shifts the price path by at most 0.06 percentage points.

Output and Bank Rate. Neither the baseline nor the adjusted shock produces an immediate contraction in real GDP (Figure 4). Under the pre-2020 full-interaction adjustment, GDP initially rises following the measured tightening, peaks at approximately 1.3 percent after one year, and crosses below zero only after about two years, reaching a trough of -0.95 percent at month 31, against -0.82 percent under the baseline. The adjustment therefore deepens the later contraction, with a narrower conditional confidence band at later horizons, but does not eliminate the initial output expansion. The Bank Rate path differs only over the first nine months.

Sources of the price and output responses. To see which component drives each response, I decompose each baseline local-projection coefficient into the contributions of the adjusted shock and of the committee component. Because the baseline shock is the sum of the two, these contributions add up exactly to the baseline response (Appendix K.2). The two responses tell different stories. For the price level, the committee component contributes positively around the window in which the baseline puzzle appears, 17 to 23 months after the shock: at the start of that window in all three adjustments, and throughout it in the full-interaction adjustment, where it accounts for much, though not all, of the baseline increase. For output, the initial expansion comes from the adjusted shock in all three adjustments; in the level and forecast-interaction adjustments the committee component contributes negatively to GDP over the first year and a half, working against the expansion rather than producing it. The committee contributions are imprecisely estimated, and their conditional confidence bands include zero. Omitted committee hawkishness therefore contributes to the price puzzle in the point estimates but is not the source of the output puzzle, whose origin lies in the adjusted shock, for example in central-bank information effects (Ricco and Savini 2025).

Sensitivity. The patterns are similar across unrestricted horizon-by-horizon projections, spline bases of seven, nine, and eleven functions, and Newey–West bandwidths of 12, 24, and 36 months; the one exception is the full-sample price difference, which is significant at bandwidths of 24 and 36 months but not 12. Appendix K.2 reports the level adjustment in both samples and the forecast-interaction adjustment before 2020, Appendix K.2.1 the sensitivity grid, and Appendix L repeats the full-sample projections excluding every outcome window that intersects the pandemic.

Comparing the paths. Appendix K.2 compares the adjusted and baseline paths directly, using both paired inference conditional on the estimated shocks and a block bootstrap that also re-estimates the generated shocks (Appendix K.2.1). Some conditional joint tests reject equality of the paths, most clearly for GDP under the full-interaction adjustment and for prices under the level adjustment, but the re-estimation bootstrap does not. I therefore read the evidence as a change in the shape of the point estimates, most clearly for prices, rather than as a precisely estimated difference between the paths; the adjustment does not by itself deliver a conventional output response.

9 Conclusion

Monetary policy is often set by committees, yet empirical policy rules typically treat the central bank as a single decision-maker. This paper asks whether committee hawkishness affects how monetary policy is set and, as a result, the monetary policy shocks used to assess the effects of policy. It turns the Bank of England's member-level voting record into a meeting-frequency measure of where the Committee stands, covering the independence period through July 2026 and using report-vintage Bank forecasts.

Members hold persistently different positions relative to their colleagues, but those positions also evolve, so committee hawkishness moves at meeting frequency mainly through new votes by sitting members, with turnover mattering over longer horizons. Committee hawkishness belongs in the policy rule. The previous meeting's level and change of average hawkishness raise adjusted R^2 by 20 percent and lower the rule's out-of-sample forecast error by 25 percent. How divided the Committee is also matters: for a given average, a more divided Committee sets looser policy.

The same information predicts the conventional monetary policy shock, accounting for up to a fifth of its variation in the full sample. Where that predictability comes from is the

paper's central result. Once the shock's own history is controlled for, the level and change of hawkishness add nothing, whereas the interaction terms, which let hawkishness change the response to the Bank's forecast, remain significant. They survive a placebo that rebuilds the measure from the shock alone and a measure that classifies votes without reference to the rule, and in the full sample they also predict part of the market-based UKMPD Target surprise. The formative-experience index, which uses no votes, enters through the same channel. What the Committee contributes to the policy rule is therefore a change in the weight the rule places on the Bank's own forecast rather than an omitted hawkish tilt, and it is this committee component that the conventional shock absorbs.

Removing the committee component changes the transmission point estimates. In the pre-2020 sample the full-interaction adjustment removes the price puzzle at every horizon, and in the full sample the level adjustment does the same. Most of the baseline price increase comes from the committee component: the conventional shock labels part of the Committee's systematic response to its inflation forecast as a surprise, the mechanism Sims (1992) proposed for the price puzzle. The output puzzle remains. Real GDP still rises for about a year after a measured tightening, and that expansion comes from the adjusted shock rather than from the committee component. I read these results as changes in the point estimates rather than as a structural decomposition of transmission.

Vote-implied hawkishness is predetermined rather than structurally exogenous and is generated from an estimated benchmark, so the exercise identifies an omitted systematic component of policy rather than a causal effect of member preferences on the decision. Because the activity and unemployment interactions are not centered, and interactions with persistent regressors may also capture some nonlinearity or gradual change in the rule, the interaction terms are best read as capturing committee-related variation in the rule's response to the outlook.

Treating a committee as a representative policymaker discards information that is systematic, observable before the decision, and consequential for what empirical work reads as a surprise. Existing shock series remain usable, but where individual votes are observed, the balance of views entering the meeting belongs to the systematic response rather than to the innovation. The approach requires only attributed votes and a policy benchmark, so it can be applied at other central banks that publish how their members vote.

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